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The permanence of friction ridge skin and persistence of friction ridge skin and impressions: A comprehensive review and new results

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ABSTRACT

This study addresses the permanence and persistence of friction ridges and the persistence of impressions made from these friction ridges over months and years. Permanence is the unchanging presence and appearance of friction ridge arrangements and their attributes between recurring observations of the skin. Permanence was evaluated from direct photographs of fingers collected over a period of 30–45 days (covering one or more skin regeneration cycles) as well as after 8 or more years had elapsed. Persistence embodies the operational concept of whether or not a pair of images displays sufficient similarity upon which to base an informed decision that they were made by the same finger, while acknowledging certain dissimilarities or distortions due to friction ridge physiology, image capture, matrix, substrate, and applied pressure. Persistence applies to both friction ridge skin and impressions made from these friction ridges. Permanence and persistence of skin were assessed from direct photographs of fingers taken two months apart and from finger photographs separated by an interval of at least 8 years. Permanence and persistence were also assessed from impressions taken over 4 months, as well as those separated by 8–53 years. Variability due to capture method was assessed by using four image capture methods over a four month period: direct photography of fingers, impressions captured by ink, holographic imaging, and live scan. Qualified latent fingerprint examiners assessed all changes observed over time, as well as any limitations imposed by capture method.

The practice of comparison and identification of fingerprint impressions was upheld, as was the prevailing use of the word persistence to describe stability of friction ridges. All photographs and impressions of the same finger were identifiable as originating from the same source. Within all the periods of observation, level 1 detail was permanent and persistent. Persistence, but not permanence, was supported for level 2 detail. Notably, the small changes observed were only in appearance; there were no changes in the presence of new, or absence of existing, minutiae. Level 3 details of ridge edge shape and pore presence were neither permanent nor persistent. Ridge width was permanent and persistent. Incipient ridges were neither permanent nor persistent.

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1. Introduction

Friction ridges on the fingers, palms, and soles of the feet remain a standard for positive identification purposes to track arrests and imprisonments, to support criminal investigations, to aid in humanitarian efforts by identifying unknown decedents in a mass disaster, and to serve as a biometric identifier in modern-day functions.

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In the first Daubert hearing on the acceptability of fingerprint evidence, the U.S. District Court took judicial notice of the permanence and uniqueness of fingerprints, i.e. that the concepts were so well accepted that they could be taken as fact. On appeal, the taking of judicial notice was judged in error, though harmless [1]. The same judge took judicial notice of the uniqueness and permanence of fingerprints in U.S. v. Llera Plaza [2]. *Uniqueness* refers to the fact that friction ridge skin is individual, not only from person to person, but from finger to finger for the same individual. The precept of uniqueness is not the focus of this study. Permanence of friction ridge detail is an underlying assumption of fingerprint identification. We define *permanence* as the unchanging presence and appearance of friction ridge arrangements and their attributes between observations of the skin. Permanence is best assessed by direct observation of the skin and may be confirmed by comparing photographs of the finger collected over time. In previous studies and when no longitudinal photos were available, impressions were used to investigate permanence, albeit indirectly. Interpretation of impressions is complicated by effects of pressure, matrix, substrate, and image capture. Due to these factors, no two impressions of the same finger are ever exactly alike.

The operational manifestation of permanence for fingerprint comparison is the concept of persistence. For comparisons of impressions, *persistence* connotes a reliable two-dimensional representation of the three-dimensional details in the skin, displaying sufficient similarity upon which to base an informed decision of whether or not two impressions or finger photos represent the same finger. For impressions, persistence acknowledges dissimilarities and distortions due to factors such as matrix, substrate, and applied pressure. An important determinant of persistence is the fidelity of transfer of ridge detail from the finger to a substrate. The level of tolerance for distortions is informed by an examiner's experience and knowledge of friction skin morphology, skin physiology, and models of pressure effects. Transfer fidelity, the degree to which an impression captures fingerprint detail without adding extraneous information or losing information of the original, is limited by variability in the condition of the skin (including physiologic state and ambient environment) and by limitations of the capture method. Transfer fidelity may be tested by comparing impressions and photos collected contemporaneously. The concept of persistence will also be applied in this paper to identification via finger-to-finger comparisons. For comparisons involving direct photographs of the skin, *persistence* connotes sufficient similarity upon which to base an informed decision of whether or not the same finger is represented. For comparisons involving either impressions or skin, persistence subsumes dissimilarities attributable to natural processes, e.g., physiological dynamics and aging.

Until fairly recently, most fingerprint examiners testified, and new examiners were trained, that fingerprints (and by inference, volar skin) are permanent and unique. These dual assertions were supported by literature on embryological and biological research, observations of impressions collected over a considerable time, and personal and collective empirical experience derived from comparison of innumerable case work impressions. In the mid-1990s, acknowledging that minor changes can occur in the skin over time, usage of the term permanence began to decline, gradually being replaced by persistence. A 2009 review of the science consistently uses the word persistence [3], including when elaborating on 2003 SWGFAST Standards for Conclusions that repeatedly refer to permanence [4]. Both terms continue to be used. For example, when referring to Galton's book *Finger Prints* (1892), The NIJ Sourcebook [5] credits Galton with establishing that "friction ridge skin was unique and persistent" (p. 1–9) but elsewhere that "fingerprints are both permanent and unique" (pp. 5–6). The current SWGFAST document refers exclusively to persistence [6]. Vanderkolk addresses the matter adroitly:

> The morphological surface structure of friction ridge skin is persistent. Often, the friction ridge arrangement (ridge flow and minutiae) has been described as permanent. However, the cellular surface of the friction ridge skin is not permanent. Surface cells are replaced on a regular basis. The competing forces of regenerating skin cells and the effort of maintaining the form and function of the organ of skin produces a persistent, *not* permanent, naturally patterned surface with all of its minute and microscopic features. In other words, the process strives to reproduce, but cannot perfectly reproduce, the patterns of the preceding cells so that the arrangements of replacement cells can follow the form and function of the replaced cells. Microscopic variations do occur. Aging of skin is an example of persistency; although patterns in friction ridge skin are not perfectly permanent, they are remarkably persistent over time.
>
> For friction ridge skin to be valuable for the examination of two prints, the unique features of ridges, creases, scars, and imperfections in the skin that had been recorded as details in two prints must be persistent between the two occurrences when each print was made. Persistency is all that is needed, not permanency [7, p. 9–4].

Thus, although nearly synonymous in everyday speech, there is an important distinction between the terms. Permanence connotes everlasting stability. Persistence implies sustained existence of a trait, but can allow slight change or a point in time when a change occurs.

Three levels of detail are used to describe fingerprints. Level 1 indicates pattern class (arch, loop, or whorl) and ridge flow. Level 2 refers to the path, direction, and termination of specific ridges, and minutiae if present. The minutest detail, including the size, shape, and relationship to one another of sweat pores and ridge edges, is referred to as level 3 [8, pp. 95–96].

This study takes a two-phased approach to assess permanence and persistence of all three levels of detail over two time periods in a controlled environment. The first phase involves capturing photos of the fingers and impressions made from the same fingers every 30–45 days, thus extending over several skin regeneration cycles [9, p. 204]. A second phase is designed to assess permanence and persistence spanning 8–53 years.

This study is designed to test two hypotheses by attempting to disprove them. The first is foundational, the second is operational:

H1. The three levels of detail that exist on the skin are permanent.

Under this hypothesis, the configuration of ridges on the skin, i.e., the arrangement and sequence of all ridges combined, would remain the same throughout life. A ridge will regenerate in the same location for each cycle and have the same ridge path (barring injury through the basal layer). Each ridge will regenerate with permanent structure in terms of ridge ending(s), dividing ridge(s), width, edge shapes, and pores.

While the biological sciences support permanence of friction ridge skin, they do not address whether the impressions made from the finger will be faithful representations when recorded on a surface. Since real-world uses of fingerprints typically involve impressions, it is important to relate photographic observations of friction ridge skin to what is seen in impressions collected at the same time. Although fingerprint examiners are unaccustomed to comparisons involving photographs of fingers, such comparisons are occasionally encountered. Cases have been noted where

"selfies" of a hand holding firearms evidence, posted on social media or recovered from a mobile telephone, were of sufficient quality to effect an identification [10]. A sexual offender was identified by magnifying his fingers that were visible in images captured in the midst of committing a sexual act with a child [11,12]. In another article, photography of a decedent's fingerprints was advocated in certain cases when it is impossible to acquire an inked print of sufficient quality for comparison [13].

The second hypothesis to be tested involves the concept of persistence:

H2. The three levels of detail remain persistent on friction ridge skin and in impressions.

The study design acknowledges three basic assumptions:

A1. Photographs of the finger represent the ground truth of friction ridge structure and ridge configuration at a specific time.

A2. The ability to determine the persistence of fingerprint details revealed in impressions depends on transfer fidelity of the capture method.

A3. Each capture technique, including photography, has inherent limitations relating to quality/clarity due to the technique itself, the skill and ability of the operator, or both.

It should be noted that all capture methods used in this study represent a three-dimensional surface in two dimensions. Ground truth is a representational image that, subject to any limitations of its own, serves as a reference for comparison with other methods. It represents "definitive knowledge of the actual source of an impression" [6]. Photography of friction ridge skin was chosen to represent ground truth because it removes the effect of pressure and does not require the use of a matrix such as ink or a substrate such as a fingerprint card.

The following rules shall be used to determine whether features are permanent (hypothesis 1) or persistent (hypothesis 2):

R1. For hypothesis 1, a feature will be deemed permanent if it appears in the same shape, size, and configuration relative to the overall pattern in two photographs from the same source.

If even one difference in any of those characteristics appears between photographs during the period of observation, the feature will be deemed not permanent. For example, if a dividing ridge appears in one photo, but appears as an ending ridge in another photo, then that feature would be deemed not permanent.

R2. For hypothesis 2, a feature will be deemed persistent if it appears in two photos or impressions from the same source with sufficient correspondence to support an identification decision.

Identification is defined as, "The determination by an examiner that there is sufficient quality and quantity of detail in agreement to conclude that two friction ridge impressions [or finger photos] originated from the same source" [14]. A feature that is present in one impression but not in the other (discrepancy), shall be deemed not persistent. For example, if an incipient ridge appears in one impression but not the other, then incipient ridges would be deemed not persistent.

In this study, the concept of permanence applies only to direct photographs of the finger, while persistence applies to both impressions and photographs of the finger. Note that the standard for permanence, since it applies to ground truth, is much higher than that for persistence. Persistence allows for variations in capture method, physiologic variation, and effects of pressure, matrix, and substrate. Many such dissimilarities are readily explained by one or more of those factors.

The initial impetus for this study came about from several challenges to the foundations of fingerprint science. The U.S. Supreme Court's decision in *Daubert v. Merrell Dow Pharmaceuticals, Inc* [15] redefined the standard for admitting expert testimony, specifically with respect to what constitutes science. Testing the premise of permanence by comparing fingerprints from the same individual over time was suggested within the court's opinion on the *Mitchell* appeal [1]. The Scientific Working Group on Friction Ridge Analysis, Study and Technology (SWGFAST) endorsed additional studies on the reproducibility (i.e., "variations among multiple impressions of the same friction ridge skin") and persistence of level 2 and level 3 detail in the skin, which "relates to variations in the same friction ridge skin over an extended period of time, considering external impacts (e.g. environmental changes, occupational exposure, aging and chemotherapy or other medical treatment)" [16]. Further, as recommended by the Office of the Inspector General's progress review of the FBI's response to recommendations related to the Brandon Mayfield case [17], this study also addresses the persistence and transfer fidelity of several fingerprint capture methods (termed "reproducibility" in that report). The original report defined reproducibility to "refer to the issue of whether level 3 details are reproduced in latent prints with sufficient persistence and reliability of appearance to provide valid individualizing power" [18, p. 196]. It recommended that the FBI "should consider shifting at least some of its emphasis on research of Level 3 detail from the issue of permanence to the issue of reproducibility and defining the circumstances under which Level 3 detail should be utilized"[18, p. 217]. The Scottish Fingerprint Inquiry Report also recommended additional research into "the weight to be given to third level detail, and as to its reliability" [19, sec. 35.134].

2. Literature review

The biological sciences support the uniqueness and persistence of the arrangement of friction ridges, their features, and the features' spatial relationship in skin of the fingers, hands, and feet. Numerous studies have demonstrated that, during gestation, the human body generates volar pads on the palmar sides of the hands and soles of the feet. Subsequent regression of volar pads and appearance of primary ridges, along with both genetic and epigenetic factors, influence the ultimate structure of each ridge and the arrangement of all ridges [20–24].

The inexorable regeneration of these ridges with the same structure and configuration resides in cell creation, proliferation, association, and migration. The three-dimensional morphology of friction ridge skin is maintained and persists in part due to the physical cell attachments throughout the layers of the skin, cell regulation, and communication. The basal layer of the epidermis, also known as the generating layer, provides the template for the surface ridges and the spaces between ridges known as furrows. The literature on the structural biology of friction ridge skin concludes that friction ridges constantly regenerate through creation of new cells at deep tips of epithelial rete (interpapillary) pegs at the basal layer [23,25,26]. While not all ridges, or even portions of ridges, regenerate simultaneously or at the same rate, the result is that approximately every 30 days the finger has gone through a complete regenerative cycle [9]. Cell communication ensures that basal cell proliferation is stimulated and inhibited in a coordinated manner. As the cells in the basal layer divide, cell-to-cell attachments and communication ensure that the cells move together.

The biological basis for persistence of friction ridges has been reviewed elsewhere [21,23,27,28]. The most salient manifestation

of persistence is the reappearance of the same arrangement of friction ridges following their unintentional or deliberate loss. Attempts to obliterate fingerprints by abrading the friction ridges will fail unless damaged to a depth of at least 1 mm, the approximate thickness of the epidermis [29–32]. Over a period of time, the friction ridges will regenerate from the basal layer. Faithful regeneration of friction ridges is further illustrated by recent reports of cancer patients who, having lost their friction ridges while under certain drug regimens, can again pass fingerprint biometric tests when drug therapy ceases [33–36]. Accurate regeneration will also occur after superficial abrasion or removal of friction ridges, either intentionally or when people involved in occupations such as manual labor, which cause wear to the epidermis, cease that activity [30,31,37–40].

Foundational studies on the persistence of friction ridge impressions and their features focused on level 1 and level 2 details [40–44].[4] Numerous demonstrations of the permanence of friction ridge impressions on the palmer surface were made by fingerprint pioneer Sir William James Herschel. He experimented by fingerprinting himself at the ages of 26, 44, and 83, as well as by collecting fingerprints over a period of many decades from a number of other individuals, finding no changes in any case [42]. German anthropologist Hermann Welcker of the University of Halle also verified the permanence of friction ridge impressions by recording his own thumb and palm prints, separated by an interval of 41 years [45]. Another fingerprint pioneer, Sir Francis Galton, showed that fingerprints were unique and unchanging from childhood to death and even after [41]. He also demonstrated that injury could lead to obliteration and permanent scarring of friction ridges. Galton took impressions from several individuals, between the ages of 2 and 80 years, separated by intervals of one to three decades. Using impressions from 15 individuals that had been collected by Herschel over periods from 9 to 15 years, Galton's general conclusions were that even though the friction skin could be altered over time, the configuration of ridges and minutiae remains the same. He cautions against attributing undue importance to minor differences between impressions that are observed at specific minutiae, "chiefly connected with the commencement of interpolated ridges," e.g., dividing to ending ridges or vice versa [44, p. 10]. Interestingly, in one pair of fingerprints from a boy taken at ages $2^{1}/_{2}$ and at 15, Galton reported 42 points of agreement but one point of disagreement: a dividing ridge was no longer present at the age of 15 [41, p. 93]. Henry Faulds also demonstrated that fingerprint impressions were both persistent and unchangeable. He detected no changes in the fingerprints collected over a period of two or more years from children, nor in adults whose ridges regenerated after being shaved or abraded away, nor in his own fingerprints over a period of 30 years [40]. In another demonstration that fingerprints do not change, even during the period of active growth, Wilder and Wentworth collected a thumb print from a child every two years between the ages of four and fourteen years [43, pp. 130–131]. A 12-year longitudinal study assessed persistence of impressions using automated matcher scores [46]. Match scores were stable over the period studied, although there was a significant decrease in true match scores as the interval between collections of the two fingerprints increased. True match scores also decreased as fingerprint quality decreased. The overall conclusion was that fingerprint recognition accuracy was stable over time.

Perception varies widely as to the reproducibility and evidentiary value of level 3 detail [47]. Two comprehensive reviews of fingerprint examinations for high profile cases have admonished that undue significance should not be ascribed to level 3 details, particularly if they are invoked primarily to confirm similarity but discounted if dissimilar—a form of confirmation bias [18,19]. Probably the earliest publication to assert the permanence of pores, as well as of the ridges on which they reside, was by Nehemiah Grew using impressions [48]. Dr. Edmond Locard was the first to publish extensively on the use of level 3 detail for identification [49,50]. His article, and others published afterwards, explained the theory of poroscopy, the size, shape, and location of pores, and how the use of pores could supplement a fingerprint comparison by lending supporting data [49]. The book, *Personal Identification* [43] is considered the first scientific research supporting level 3 detail as unchanging and unique [51]. Its authors note that pores can vary in size, shape, location, and frequency, supporting Locard's view that pores can aid, or in some cases be used independently, to effect an identification. Pores in inked impressions collected over intervals as long as 48 years, although quite variable in size, shape, and individual appearance, were shown to retain their spatial position relative to one another [52]. Several papers have advocated use of pore area for comparison [8,53–56], but other studies concluded that pore area is too variable to be useful. Most fingerprint examiners consider pores non-reproducible [47]. Although one study found pore area to be unchanged in high quality inked prints [57], others did not [58,59]. Pore area was also reported to be non-reproducible in both latent impressions [57,60,61] and live scans [62]. A study involving 516 live scan images of 10 different fingerprints determined that specific pores were visible in 91% of the images on average, with the least reliable pore visible in 75% of the images [57]. Using direct photographs of the finger, pore area was observed to be reproducible over a period of one hour but not reproducible over a period of one month [62]. A recent study quantified the weight that might be assigned to observed agreement or disagreement of spatial arrangement of pores [56]. It reported that, based on a model using four small areas including an ending ridge and associated pores, false inclusions occurred at a rate between 4–8% and false exclusions occurred between 10–23% of cases. An example of persistence of pores over a period of ten years was offered in the Daubert hearing for the *U.S. v. Mitchell* case [63]. Two other studies, involving repeated photographs of one finger while being pressed against glass with different levels of pressure, reported high reproducibility in regard to presence or absence of pores but weak reproducibility in their size, shape, and lateral position [61,64]. Faulds noted of pores that, "they remain locally fixed in position" [but] " . . . their physiological activity being very variable, their shapes are constantly altering. They may be nearly closed one moment, and quite patent the next . . . " [65]. A model of the temporal variability in the visibility of pores proposed that pores are like valves, opening when sufficient sweat accumulates within the sweat ducts to overcome the resistance of the topmost stratum corneum (horny layer) [66–68]. This model was partially validated by dynamic optical coherence tomography (Fig. 1), demonstrating opening of a closed pore on a human finger under physiological conditions [69]. Those investigators observed variation over time in pore diameter, amount of fluid in the lumen of a sweat duct, and slower response to stress due to a learning effect of the sympathetic nerve controlling the sweat gland [70]. The same team reported that sometimes a sweat pore fails to open, even when the stored sweat in the ductal lumen is at maximum level, so-called internal sweating [71]. Variation in secretory activity of individual pores has been demonstrated with a hydrochromic sensor [72,73].

The term, "edgeoscopy," referring to the morphological attributes of friction ridges, is attributed to Salil Chatterjee [74]. He classified the edges into seven most commonly observed ridge shapes. Ridge edge shape and width sometimes correspond in a

[4] The term persistence has been substituted if permanence was used in previous work to describe the sustained existence of friction ridge detail.

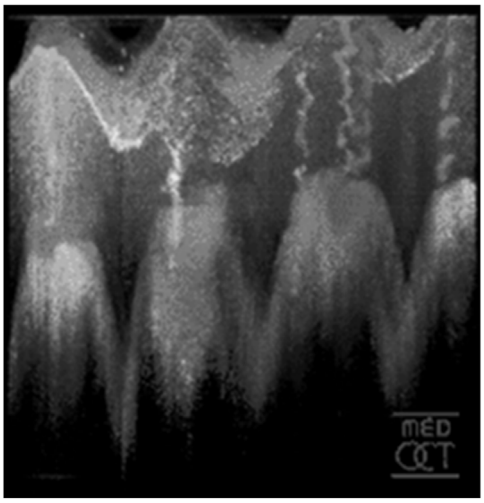


Fig. 1. Optical Coherence Tomography (OCT) tomogram, depicting stratum corneum on top and stratum lucidum (connection to stratum spinosum) in the middle. At the bottom are superficial parts of the dermis. Sweat ducts are clearly visible [medOCT 2005].

comparison and may therefore support a decision, but being highly dependent on pressure and development method, are typically not well reproduced [8,61,64]. In mated impressions of 100 individuals, separated by a duration between 5–48 years, matching edge features were found in 75 pairs, with a total of 309 features detected [52]. The author did not attribute this observation to lack of persistence, but to susceptibility of the ridge edges to deformation and damage. The number of edge features also decreased with increasing duration between collections. Ridge spacing (furrow width) variation is possibly attributable to physiological occurrences such as weight gain/loss, ambient temperature, or variations in degree of hydration of the stratum corneum [75,76]. As Galton observed, "the pattern grows simultaneously with the finger, and its proportions vary with its fatness, leanness, usage, gouty deformation, or age" [41, p. 98].

Incipient ridges are variously known as immature, rudimentary, nascent, interstitial, subsidiary, extraneous, false, or interpapillary ridges or lines. Incipient ridges, being thinner and lower than ordinary ridges, may or may not be represented in impressions, nor do they contain pores [77,78]. Incipient ridges, which have characteristics of both level 2 and level 3, may be used to support a conclusion in the friction ridge examination process [79,80]. An identification based solely on incipient ridges was published by Reneau [81]. Wilder and Wentworth detected no changes in incipient ridges in inked impressions collected every two years from a child between the ages of four and fourteen [43].

As skin ages, it undergoes numerous changes in morphology and physiology that may be manifested in surface appearance, fingerprint impressions, and success in development of latent fingerprints. Galton observed that, ridges " . . . continue to be sharply defined until old age has set in, when an incipient disintegration of the texture of the skin spoils, and may largely obliterate them . . . " [41, p. 58]. The structural support for the protrusion of friction ridges above the surface is provided by the dermal papillae [78], which are stabilized by interdigitating rete (interpapillary) pegs of the epidermis [82]. With age, both the rete pegs and the dermal papillae progressively disappear [83,84]. The dermal papillae flatten, decrease in number, and become wider [27,85–89]. The junction between dermis and epidermis becomes flatter with age and the thickness of the dermis decreases markedly, while that of the epidermis remains essentially unchanged [84,90–95]. The consequence of these age-related changes is the "flattening" of friction ridges familiar to fingerprint examiners [27,78,96]. Due to loss of collagen and elastic fibers, coupled with the destabilization of the dermal-epidermal junction due to flattening of the dermal papillae and rete pegs, the skin becomes looser, prone to wrinkling and epidermal gliding and shearing, potentially resulting in poorer impressions and a decrease in fingerprint quality [27,83,88,90,96–99]. Age-dependent flattening of friction ridges may be due in part to increasing uniformity of dermis-epidermis adhesion in ridges and furrows, which was in turn inferred from an observed increase in the number of small papillae with age [100]. These changes in the character of fingerprints have been exploited for automated age estimation of the donor [101,102]. Ridge density tends to decrease with aging, correlated with increased hand size [96,103]. Based on the observation that the quantity of incipient ridges tends to increase among groups of people of greater age, it has been inferred that incipient ridges may continue to develop and manifest themselves over the lifetime of an individual [43,104,105]. Pores have been reported to increase in size with age, but not in number [106].

The quantity and composition of fingerprint residues also changes with age. The number of eccrine glands decreases with resultant reduction in secretions [88,94,107]. Sebaceous gland secretion also diminishes, with age-related variations in fatty acid composition between the sexes and among different ethnic groups [108–115]. These glandular changes affect image quality attainable with various sensors [116] and likely, latent fingerprint development outcome. True match scores from an automated system have been reported to decrease as the age of the subject increases [46,97]. Poor image quality of impressions taken from older people leads to an increased rate of failure to enroll in a biometric system [96,117]. In 2015, the rejection rate of fingerprints submitted to the FBI's Next Generation Identification (NGI) system remained at approximately 1% up to age 45 and steadily increased with age thereafter, especially after age 60. At about age 80, the rate reached a peak of 28%, then began to decline (J. Stout, personal communication, October 19, 2015).

Creases, wounds, and scars are not themselves friction ridges, but they can affect the appearance of impressions, aid in orientation, and support the comparison process [79,118,119]. Flexion creases, found near the joints of the hands, fingers, and feet, lack ridge structure and interrupt ridge flow [120]. Flexion creases first appear during embryological development [121,122] and, with two minor exceptions [123], they develop independently of flexion movements [121,122,124–127]. Flexion creases are strongly attached to the underlying dermal fascia [128]. Although changes in fine details were documented to occur in the distal palmar crease, the proximal and thenar creases were stable [129]. Flexion creases may be used to support an identification [45,130]. Other skin creases are less firmly attached to the underlying fascia. These secondary creases are not permanent [131–133]. Secondary creases may arise from several mechanisms, but all are manifested as white lines in impressions. White lines do not disrupt ridge flow [134] and while they may be useful for orientation, are ignored in fingerprint examination [132]. White lines have been attributed to buckling in the skin [30]. When the skin folds, the ridges in the crease are lower, and thus may fail to print, resulting in a white line devoid of information [133]. Some white lines may arise due to natural separations in the network of collagen fibers of the reticular layer of the dermis, known as cleavage lines, tension creases, or Langer's lines [128]. Although observed to be stable within a single month, changes in the presence and appearance of secondary creases were demonstrated over a period of approximately six months [135]. White lines occur more frequently in women, in certain occupations, in later life, on the middle, ring, and little finger, and in certain neurological and dermatological diseases [30,45,96,131,132,136–141]. A reported correlation of

white lines with celiac disease [142] is widely quoted in websites for sufferers of the disease but was not substantiated by other researchers [143,144]. Several dermatological diseases such as dermatitis, eczema, and psoriasis may cause temporary scaling, wrinkling and increased white lines, or loss of ridge detail [137,145–148]. Scars damage the original structure of the skin and are characterized by unnatural disturbance of ridge flow [133]. Although not extensively studied, permanence of fully-healed scars has been demonstrated and attributed to the same mechanisms that explain persistence of friction ridges [119,133].

3. Materials and methods

3.1. Sample collection over a period of 1–4 months (short term), analysis, and comparison

Fingerprint impressions of 86 individuals were recorded in April, May, June, and August, of the same year (Table 1). Photographs of fingers were recorded in April and June. All study participants signed human subject release forms approved by the FBI Institutional Review Board and were assigned a code. Fingers were inspected before image capture to insure that the fingers were free from debris. Four image capture methods were used: photography, ink, holographic imaging, and live scan. All images were oriented in the upright position. Each of the four recording sessions involved at least two of the capture techniques (Table 2).

Study participants and individual fingers were selected to achieve approximately equal numbers of each pattern classification while maintaining a balanced distribution of sex, age, and finger position. Various factors precluded obtaining a complete set of images from every participant. Complete sets representing every capture method used in each month were available for 123 of the 167 fingers studied.

Digital photographs of the fingers were captured in the April and June sessions to represent ground truth. During the April capture session, a Nikon D1H camera with a 2.6MP resolution was used. Photographs of the fingers during the June capture session were taken with a Nikon D2H camera, which has a 4.2MP resolution. In both sessions, oblique lighting was used. The photos were reversed in position left to right to facilitate comparison to other capture methods. For a group of 80 research participants, photographs from 152 fingers were collected to assess short-term permanence of friction ridge detail.

Table 1
Gender and age distribution.

(short term study)		
Age	Female	Male
25-30	15	6
31-40	10	9
41-50	12	7
50+	11	16
Total	48	38

Table 2
Data acquired during each capture session.

Capture session	Capture technique[a]	Number of participants	Number of fingers	Number of images
April	I, H, P	86	163	429
May	I, H	82	161	322
June	I, H, P	80	162	486
August	I, H, LS	68	137	411
			Total	1648

[a] I = Ink, H=Holographic imaging, P = Photography, LS = Live scan.

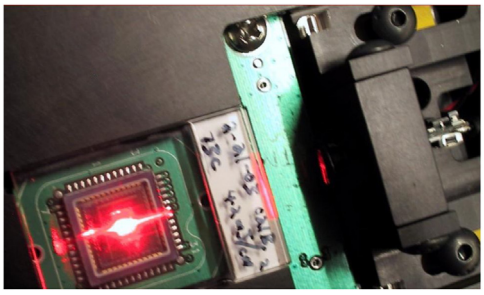


Fig. 2. Aprilis HoloSensor holographic sensor.

During all four sessions, a prototype holographic imaging station (HoloSensor, Aprilis Inc; company later acquired by Dow Corning, renamed DCE Aprilis, then STX Aprilis) was used to capture the images at approximately 4200 ppi (Fig. 2). At this resolution, level 3 details, including the size of pores, pore shape, ridge edge shape, and their relationship to one another, were usually visible. Each participant touched a finger on the glass platen and a fingerprint expert recorded the image of the central area. Between each image capture, the glass platen was cleaned with isopropyl alcohol. If necessary, the finger was also cleaned with isopropyl alcohol. The imaging area of the holographic imager was limited to a half inch square. Therefore, this imaging area served as the limit on the area compared. Depending on the size of the finger, it was not always possible to capture the entire core area. If the finger was too large to capture the entire core area, the same area of friction ridge skin was still captured from one session to the next. Before the image was saved, the exposure of the holographic imager was checked to insure good contrast in the image. A cursory assessment was conducted for factors affecting the quality of the image: deposition pressure on the glass platen, excess perspiration on the friction ridges, and movement of the finger on the glass platen.

Live scan images were recorded at 500 ppi resolution in the August session using the ID 1000 scanner that was in common use at the time when the images were acquired and was a component of the Joint Automated Booking Station [149]. Available operator adjustments include prism position and finger guide spacing to accommodate differences in finger size. The system prompts the operator to apply each of the subject's fingers to the platen successively. The platen was cleaned with isopropyl alcohol as necessary. The scan was saved after the operator indicated that it was of acceptable quality.

The standard inking process was used in each of the four capture sessions. Because any residual ink on the fingers could have interfered with the other capture methods, this method was done last. Both rolled and plain impressions of the specified fingers were recorded multiple times on standard fingerprint card stock. All cards were subsequently scanned at 1200 ppi resolution.

A first round of assessments was conducted by seven co-authors representing state and local U.S. law enforcement agencies. These persons reviewed sets of images from all four sessions of recordings. All images collected for the 167 selected fingers were analyzed and compared; 152 of these were examined by at least two assessors in the first round of examinations, with 43 reviewed by more than two assessors. Over 20,000 pairwise comparisons were conducted in the first round. In a second round, these assessments were extensively reviewed by FBI personnel.

The assessors were made aware that all comparisons involved photos or impressions from the same source (i.e., mated pairs). Assessment was confined to a pre-designated area of interest (AOI) of each finger's core, selected such that each image included

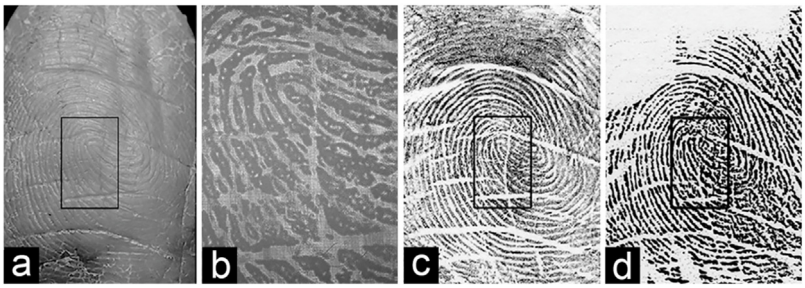


Fig. 3. Variability attributable to cracked, creased, and abraded skin over a short term within the same individual at age 55: (a) April photo; (b) May holographic (core region magnified), (c) May ink rolled, (d) August live scan. Area of interest (AOI) common to all images is indicated.

approximately the same area. The AOI for all other capture methods was limited to the area that was captured by the holographic imager (Fig. 3).

For each image pair, the assessors analyzed the degree of similarity of features within the AOI. The assessors performed a qualitative assessment first, followed by an assessment of the three levels of detail, generally in order starting with level 1 detail. The assessors were examining each pair of images in an effort to determine the extent of agreement and disagreement of any features within the AOI. All comparisons were conducted on a video monitor, using Adobe Photoshop CS4 (Adobe Systems, Inc., San Jose, CA) for side-by-side display. They documented their observations on the pairwise comparisons in a free-form matrix. These observations were used to guide reexamination of the images by FBI latent fingerprint specialists. Overall conclusions were subsequently tabulated using the comparison codes shown in Table 3. Determination of permanence or persistence was made from these results.

Several aspects of this study that could induce bias were considered. First, since all comparisons were known mates, there could be a tendency to seek similarities rather than differences. Second, one might dismiss a distortion or problem rather than seek a valid explanation. Third, past training and experience has inculcated the concept of permanence for each assessor. Fourth and most crucial, any indication that persistence does not exist could potentially undermine the entire discipline. These potential biases were addressed in several ways. All assessors were very aware of the goals of the study. Accordingly, they made a deliberate effort to detect and record all variations. Multiple assessors reviewed photographs and impressions, and they were encouraged to confer. One of the second-round reviewers (KM), although an experienced forensic scientist, is not a fingerprint examiner, so is less threatened by a potential overturning of dogma.

3.2. Sample collection over a period of 8 or more years (long term), analysis, and comparison

Prior to beginning FBI employment, inked fingerprints are collected from all prospective applicants. Thus, long-term employees are a ready source of data for an investigation of any changes in the fingerprint over an extended period of time. Employees with eight or more years' tenure were selected using the Bureau's personnel management system. For convenience in acquiring new impressions, the search was further limited to include only those employees currently assigned to the FBI Laboratory, Quantico, VA. The pre-employment inked 10-print cards of those employees were obtained and the Level I classification (loop, whorl, arch) of each finger was tabulated. Approximately 35 individuals with one or more arches (which tend to be less common) were selected to participate. An additional 165 individuals were randomly selected. The persons whose prints were selected were contacted to request their participation and to schedule a time to give a new set of inked prints. Each donor signed an informed consent form.

National Crime Information Center (NCIC) classifications [150] were independently determined for the new and old 10-print cards from each individual. When searching NCIC fingerprint records, windows are generally widened to include deviations of plus or minus one ridge count, and for arches, to include both plain and

Table 3
Comparison codes.

Level 1 changes	Level 2 changes	Level 3 changes	Comments and explanations
Creases	Ridges	Pores	Overall Conclusion
1-Appear	1-Length change	1-Better visibility	1-Ident
2-Disappear	2-Short ridge(s) appear	2-Poorer visibility	2-Non-ident
3-Size change	3-Short ridge(s) disappear	3-Linear position	3-Inconclusive
4-Other	4-Linear break(s) appear*	4-Centering on ridge	
Scars	5-Linear break(s) disappear	5-Shape	
1-Appear	6-Ending to dividing	6-Size	Common Characteristics
2-Disappear	7-Ending to dot(s)	7-Frequency	1-Over/under inking
3-Size change	8-Other	8-Other	2-Smearing
4-Other	*not due to creases	Edge Shape	3-Light/heavy pressure
	Dividing Ridges	1-Changed	4-Limited area
	1-Dividing to ending	2-Better visibility	5-Poor ridges
	2-Other	3-Poorer visibility	
	Dots	4-Other	Free-form comments
	1-Appear	Ridge Width	
	2-Disappear	1-Thicker	
	3-To ridge ending(s)	2-Thinner	
	4-Other	3-Both	
		4-Other	

*Linear breaks could also occur due to the presence of creases.

tented subtypes (codes AA and TT, respectively). The same convention was followed in locating potential differences between new and old prints, i.e., to include differences in ridge count greater than one, and to exclude plain v. tented arch differences. A total of 1050 fingers from old and new inked 10-print impressions of 105 individuals were classified. The result was that 75 finger pairs differed in ridge count by more than one.

An additional 69 finger pairs showing a single difference in classification were also examined for level 2 and 3 detail. This additional selection was adopted to increase the likelihood of discovering differences, but it is imperfect: level 2 or 3 differences could occur despite level 1 characteristics being unchanged. Thus, these additional pairs were randomly chosen to avoid potential bias of examining only fingers that differed by more than one ridge count.

For a group of 23 research participants whose fingers had been photographed eight or more years earlier, new photographs were collected from 32 fingers in order to assess longer-term permanence of friction ridge detail. For this group, a Canon EOS 5D Mark II camera with 19.5MP resolution was used. Each person signed a new informed consent form. In order to enhance visualization of skin details, some of the subjects lightly touched an ink pad prior to photography.

As in the short term study, FBI latent fingerprint examiners analyzed and compared each pair of finger photos or impressions for level 1, 2, and 3 detail. They assessed pattern type, quality represented, any differences in scars or creases, and whether there was any difference between the two prints that would affect an identification decision. All changes were defined relative to the older print and all observations were tabulated in a spreadsheet using the comparison codes in Table 3. More than a single reason could be entered in each category, as well as further comments. In addition, there were codes for the overall conclusion, common characteristics, and additional space for free-form comments. All fingerprint pairs for which a difference was noted were reexamined by one or more additional examiners.

4. Results and discussion

4.1. Variability attributable to the skin

Image quality varied in part due to the condition of each individual's friction ridge skin. The assessors noted any recent damage to the ridges. In some cases, the participants had very fine ridges that were difficult to record. Other participants had a lot of secondary creases (hereafter called simply "creases") and resultant white lines that also affected the quality of the image captured. The finger in Fig. 3 is such an example, although impressions captured by three techniques even show incipient ridges. Despite the skin appearing to be dry and cracked, the holographic imager adequately captured level 3 detail.

4.2. Variability attributable to capture method

All assessments of permanence and persistence, and indeed, analysis and comparison of fingerprints, necessitate due consideration of concomitant limitations of the capture method. Every method offers advantages and disadvantages. All impression methods are subject to variation and distortion due to pressure [79,151] and potentially, intrinsic factors such as moisture content and shallow ridges [152]. Level 3 detail was most affected. Contrast variation was an issue for all. Photographs were low contrast and often with limited depth of field. Quality assessment for each image of a fingerprint determined how much of the AOI could be examined and which levels of detail could be examined.

Live scan technology produces immediate digital images and lends itself to rapid collection of fingerprints in the field. The majority of fingerprints submitted to the FBI are now collected by this method. However, live scan image capture often failed to show the degree of detail revealed by the other methods, particularly level 3. In rare instances, the method produced potentially misleading information. Even though the inked impression (Fig. 4a) can be identified to the live scan (Fig. 4b), the ridge

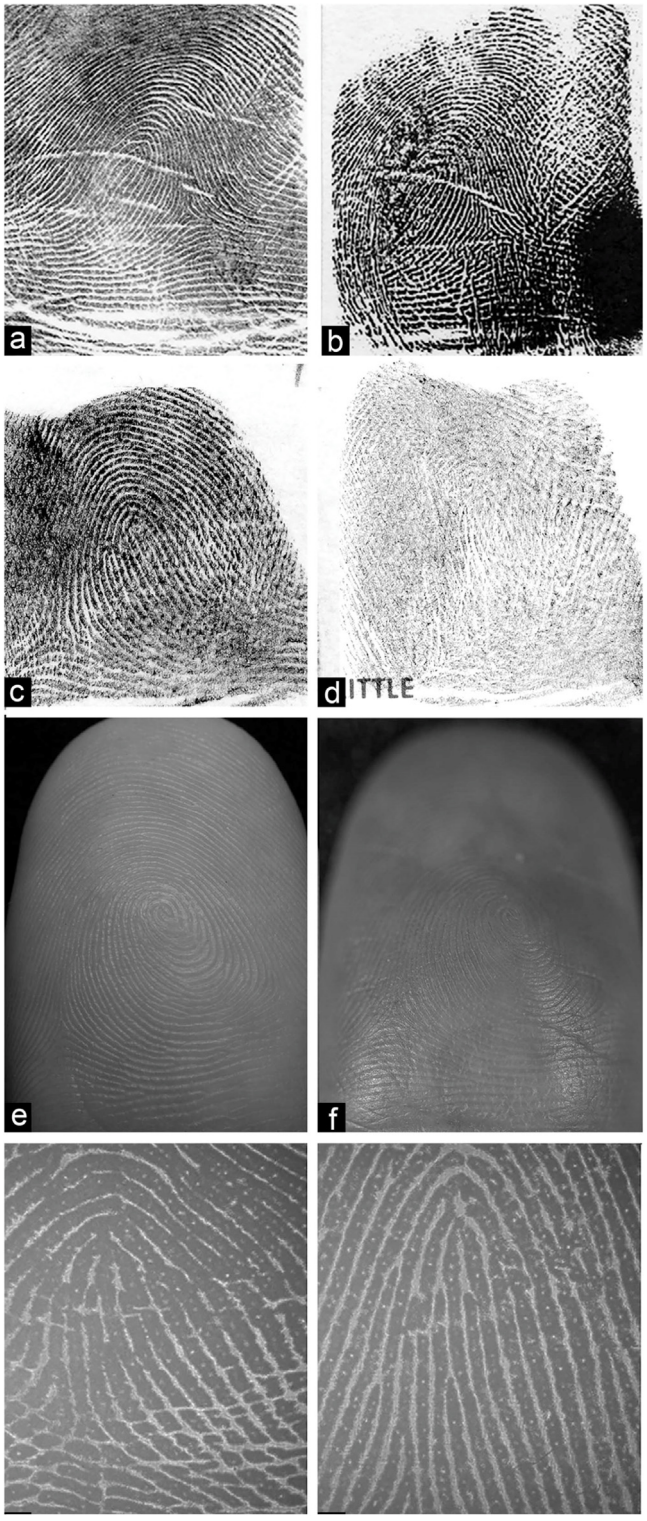


Fig. 4. Fingerprints showing variability attributable to capture method: (a,b) contemporaneous mated ink rolled vs. live scan at age 32; (c,d) mated ink rolled at ages 26 and 47; (e,f) mated skin photos at ages 44 and 53; (g,h) mated impressions captured holographically, separated by 1 month at age 30.

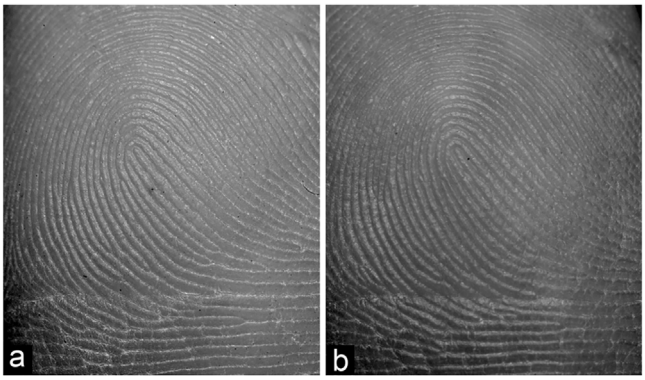


Fig. 5. Permanence and persistence of level 1 detail of skin over a short term: (a,b) mates separated by a 2-month interval at age 40.

flow is distorted in the live scan, where the left slant loop appears almost like a whorl.

Capture of fingerprints using printer's ink is the *de facto* standard. The materials are low cost and the method produces an archival paper record that is immune to changes in technology. Generally, ink capture involves using an ink pad or a layer of black printer's ink rolled onto a smooth surface and is therefore considered to be an inconvenient process. Storage and indexing of inked cards is a major drawback and they must be digitized before the fingerprints they contain can be entered into modern databases. Over- or under-inking affects the contrast and over inking can cause smearing [79,152]. Quality of ink capture was variable and was much more subject to smearing than other capture methods. The impression shown in Fig. 4d is badly under-inked and only the top half is of value for comparison.

Photography is subject to limited field of view due to depth of focus and strong dependence on lighting (e.g., various areas in Fig. 4e and f). Conventional photography cannot capture the entire finger to include its sides, as does a rolled impression. The image contrast was typically low and ridges and furrows were inverted relative to impressions, making comparison difficult. It is for this reason that some fingers were lightly ink coated prior to photography.

The principal advantage of holographic capture is its high resolution. Ridge edge shape and pores were often exceptionally well represented. But extremely high resolution came at the cost of extremely limited field of view. The small field of view made it challenging to position the core area consistently. A few issues seemed to be inherent to this type of imaging. Day-to-day variation in pore depiction has been reported using a holographic imager [153]. Ridges tended to be interrupted more frequently than in the other methods.

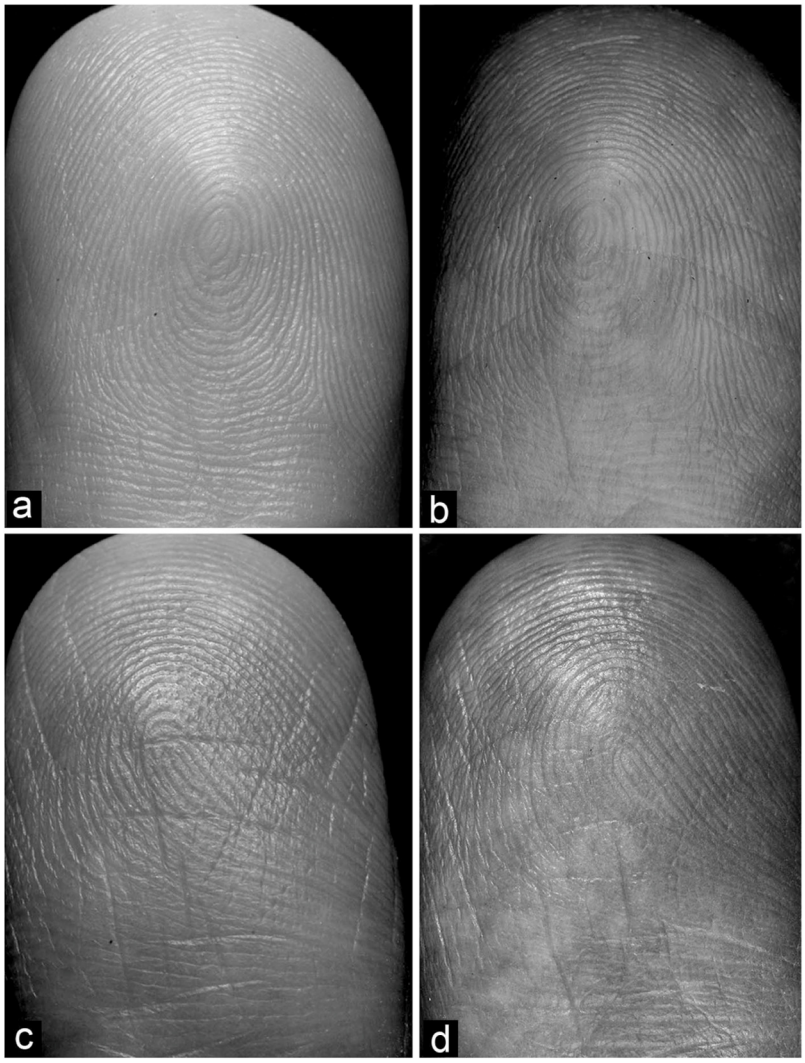


Fig. 6. Permanence and persistence of level 1 detail of skin over a long term: (a,b) mated skin photos at ages 47 and 55. Lack of permanence of creases: (c,d) mated skin photos at ages 56 and 64.

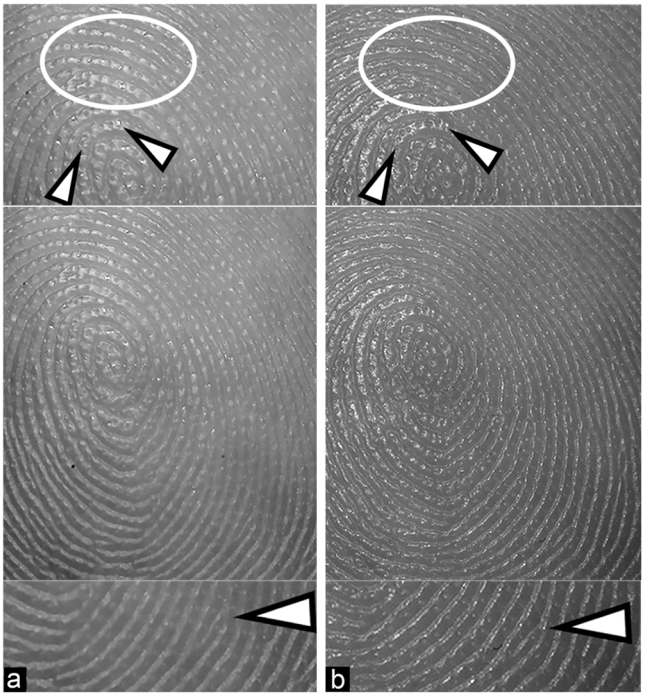


Fig. 7. Permanence and persistence of level 1 detail of skin over a long term. Persistence, but lack of permanence, of level 2 and 3 detail: (a,b) mated skin photos at ages 30 and 40, with magnified areas above and below the central photographs. In (b), the core area is somewhat more elongated; there are more ridge breaks below and to the left of the core; a dividing ridge has become an ending ridge (lower arrow); and there are variations in pore size and shape (upper arrows and circled area).

Some of the holographic scans had a canvas-like background appearance. Fig. 4g and h present an example where the holographic imager affected the level 1 ridge flow. In the April holographic scan the core appears wider and the loop appears more curved than in the May scan (Fig. 4g and h), although the core in the April and May ink rolled impressions was similar (data not shown). The ridges were wider in the April holographic (Fig. 4g) than in the May holographic (Fig. 4h), but were unchanged in the ink rolled impressions. Another anomaly sometimes introduced by the holographic imager was the appearance of artefactual lines positioned in the center of the furrows; this was not observed in any of the other capture methods. This artifact was present despite the high quality of the image, clearly displaying ridge edges and pores.

4.3. Appearance of skin over time

Due to deficiencies in focus and/or lighting, it was often difficult to locate and document variations. Generally, the appearance of fingers belonging to older persons tended to differ from those of younger persons, leading to observations of variations noted over decades that were not noted in photographs separated by only a few months. Aged fingers often showed cracking, drying, and peeling. Ridges seemed flatter and less plump with increasing age.

4.3.1. Level 1 observations over a short term

In all the direct photographs of fingers, level 1 ridge flow was permanent and persistent over a short-term period (Fig. 5).

Both H1 and H2 were supported in observations of level 1 details in the skin over a short term.

4.3.2. Level 1 observations over a long term (plus creases)

Level 1 ridge flow was permanent and persistent over a long-term period (Figs. 6, 7). Creases were not permanent, showing considerable variation. In a sample of 153 finger pairs, 77% had creases, some of which appeared or disappeared over time at approximately the same rate (32% and 27% respectively, not significantly different, $p > 0.05$). In 10% of the cases, white lines appeared and disappeared in the same finger. Fig. 6b shows an increase in creases and worn friction skin ridges below the core. In Fig. 6d, some creases are no longer apparent. Below the core, ridges seem worn but vertical creases remain. Other creases to the right of the core are less pronounced. The core of Fig. 7a has become slightly more elongated in the image captured ten years later.

Both H1 and H2 were supported in observations of level 1 details in the skin over a long term.

4.3.3. Level 2 observations over a short term

Level 2 detail was permanent and persistent in all photographs over a short-term period, of which Fig. 5 is representative.

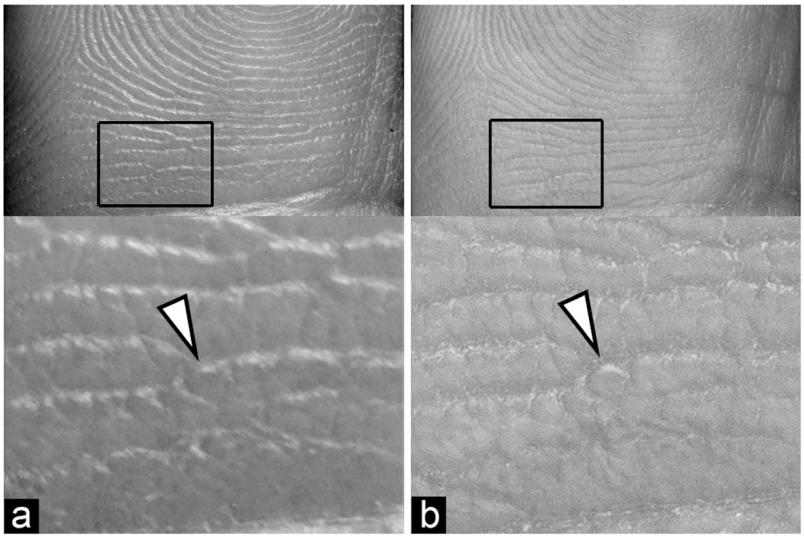


Fig. 8. Persistence, but lack of permanence, of level 2 detail of skin over a long term: (a,b) mated skin photos at ages 53 and 61. The magnified areas are indicated by boxes. The arrow indicates where an ending ridge in (a) appears as a dot in (b).

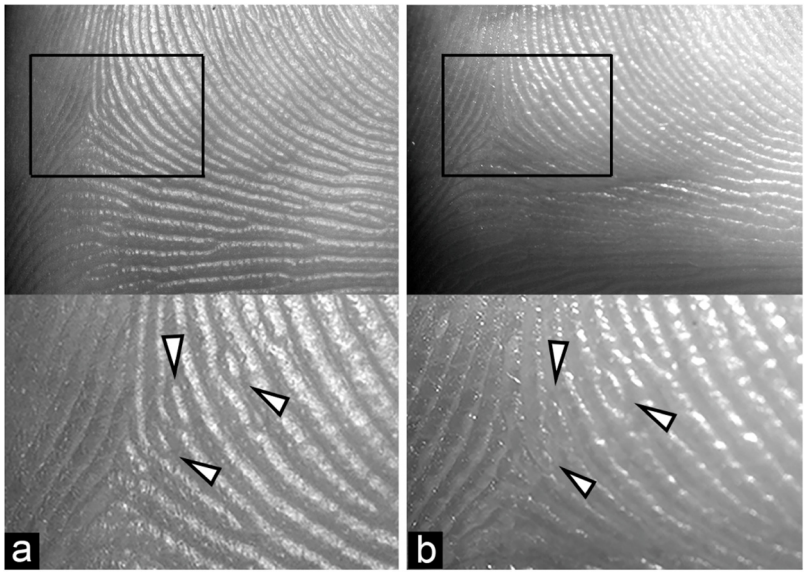


Fig. 9. Persistence, but lack of permanence, of level 2 detail of skin over a long term: (a,b) mated skin photos at ages 28 and 37. Minutiae counts differ in the magnified areas indicated. The short ridge indicated by the arrow on the right side of both images persists. The two arrows on the left indicate a short ridge visible in (a) but not in (b).

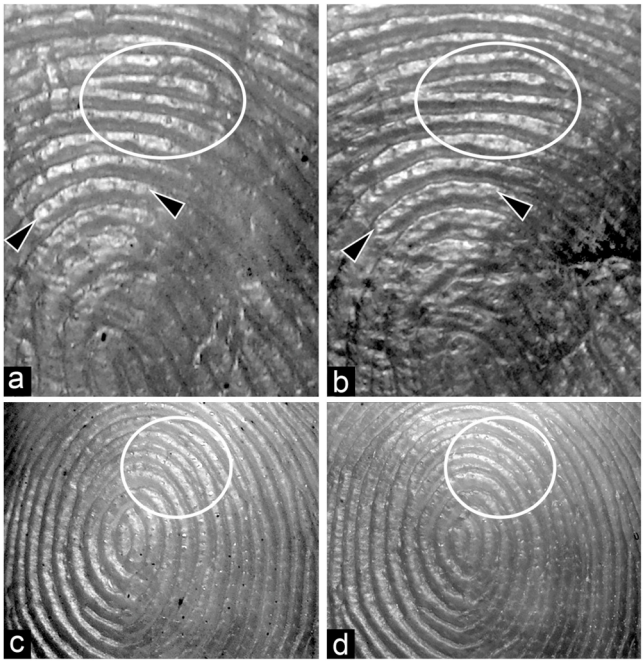


Fig. 10. Lack of both permanence and persistence in level 3 detail of skin over a short term: (a,b) are mates separated by a 2-month interval at age 54. Pore presence and location are similar on the ridge marked by arrows. Pores and ridge width are variable in the circled areas.; (c,d) incipient ridges in mates separated by a 2-month interval at age 27 are variable in the circled areas.

Both H1 and H2 were supported in observations of level 2 details in the skin over a short term.

4.3.4. Level 2 variations over a long term

Level 2 details were persistent, but not permanent, over a long-term period. A few subtle variations in appearance were observed, but all were within operational tolerance of changes associated with age and condition of the skin, including creases, cracking, and scaling.

Corresponding minutiae were annotated within common regions of interest for 32 fingers photographed over a long term. Approximately 1900 paired minutiae were counted for these 32 fingers. Only two fingers were found to have differing minutia counts (0.1%). Differences in appearance may occur even when minutia counts are the same—approximately 12 of the 32 (37.5%) had changes in the appearance of level 2 features. Most of these differences may be due to conditions of skin or capture method (lighting) but in three cases the observations are less easily explained (Figs. 7–9).

None of these variations were sufficient to alter an identification decision, i.e., all level 2 details were persistent. In Fig. 7, the lower arrow indicates where a dividing ridge in Fig. 7a appears as an ending ridge in Fig. 7b. Both to the left of the core and in the area below it, Fig. 7b shows an extensive increase in ridge breaks. Fig. 8a shows an ending ridge that appears as a dot in the later photo (arrow).

Fig. 9 denotes an area which resulted in differing minutia counts between the older and more recent capture. Although still identifiable, the area around the delta has notable variation in appearance of level 2 detail. On the right side of both images, the arrow pointing up and to the left shows a short ridge. Approximately two intervening ridges to the left is an arrow pointing down at an ending ridge. In the earlier image, that ridge appears disrupted while it does not in the later image. In the earlier image, the arrow pointing up and to the left shows a short ridge but in the later image the ridge path appears continuous. The changes in the appearance of minutia disprove the hypothesis even though there were no changes in the presence of new or absence of existing minutia that would alter an identification or exclusion decision (is persistent).

H1 was not fully supported in observations of level 2 details in the skin over a long term due to changes in appearance. Notably, no minutiae were observed to appear or disappear. H2 was supported in observations of level 2 details in the skin over a long term.

4.3.5. Level 3 variations over a short term (plus incipient ridges)

Not all level 3 features were permanent or persistent in skin over a short-term period, even though level 3 detail is sometimes used to support a conclusion in fingerprint examination. This is partly because it is the minutest level of detail in friction ridge skin.

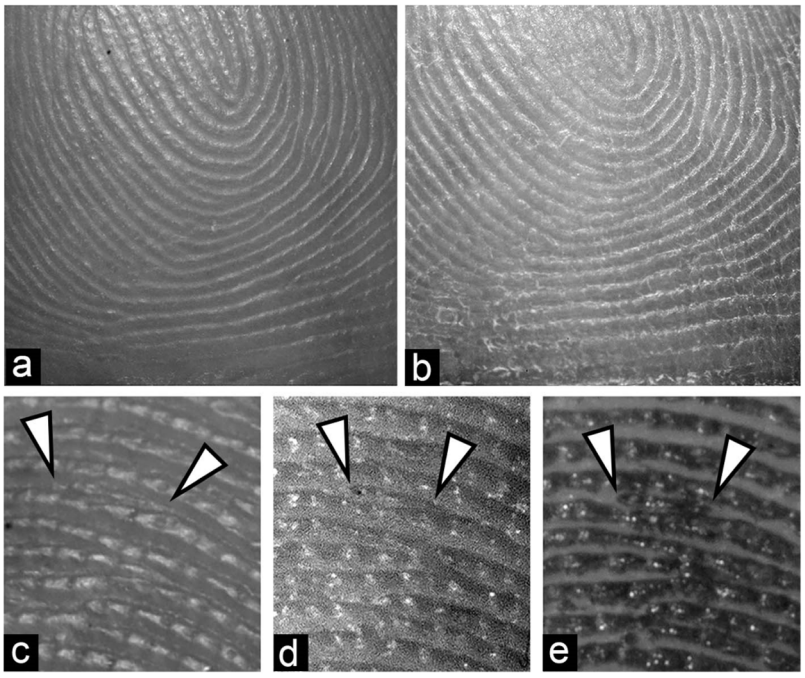


Fig. 11. Lack of both permanence and persistence in level 3 detail of skin over a long term: (a,b) mated skin photos at ages 44 and 52 show variations in ridge width, edge shape, and incipient ridges; (c,d) mated skin photos at ages 30 and 40 show variability in ridge width and shape and in incipient ridges (arrows); (e) contemporaneous and mated with d, but enhanced by touching the finger to a stamp pad.

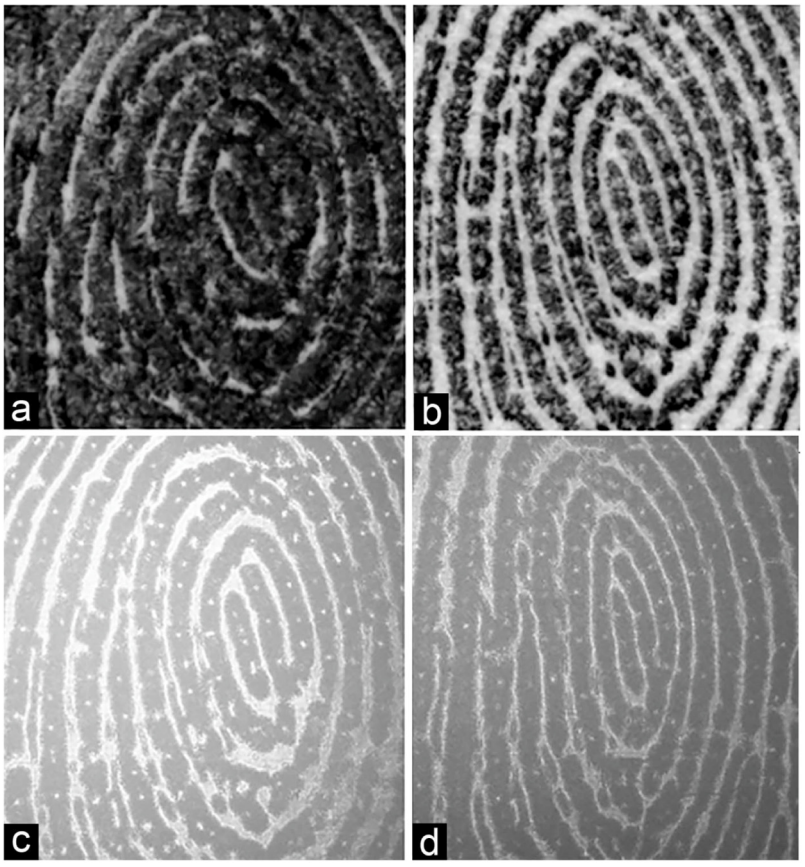


Fig. 12. Persistence of level 1 detail of impressions over a short term showing core elongation in the same individual: (a) April ink rolled; (b) May ink rolled; (c) April holographic; (d) June holographic, all at age 30. Relative to (a), the core area is elongated in the other images.

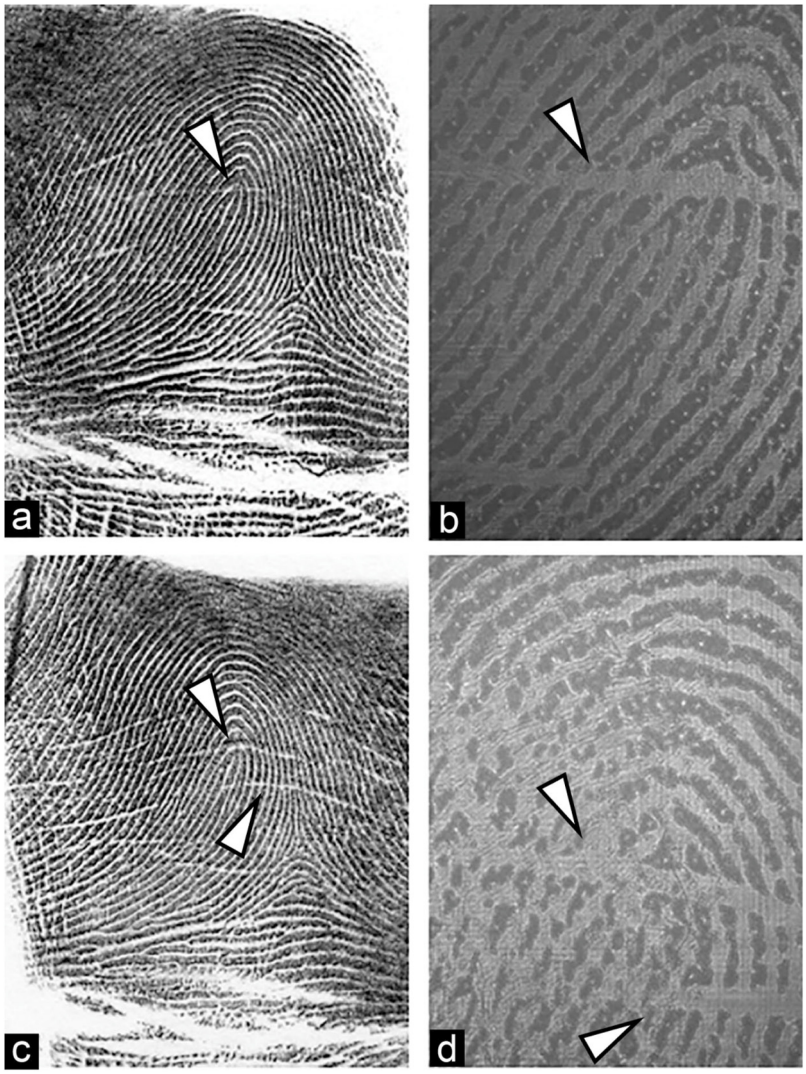


Fig. 13. Persistence of level 1 detail of impressions over a short term, but variation in creases/white lines in the same individual (arrows): (a) April ink rolled; (b) April holographic; (c) June ink rolled; (d) June holographic, all at age 29.

Ridge width and edge shape were permanent and persistent over a short-term period. Since ridge spacing variations were observed in sequential photographs over a period of minutes, that attribute was deemed persistent but not permanent.

The size, shape, and presence of pores were neither permanent nor persistent. Image quality of the photographs was less than optimal, due to low/variable contrast and focus variation across the rounded finger. Since the visibility of pores was inadequate in many of the photos, examples were selected from the approximately 15% of paired photos taken in April and June exhibiting sufficient quality to assess pores. In Fig. 10a and b, the arrows illustrate some of the pores whose presence and location did not vary, while ovals indicate regions where pores were variable. Even though pores indicated by arrows were observed in the same location along a ridge, their size and shape varied. There are several possible reasons for this observation: limitations of photography cited previously, presence of sweat, condition of the skin of the finger (e.g., dryness or creases), or physiological explanations discussed above (pores opening and closing). In such cases, the presence of pores can only be inferred, as their true shape and location is obscured. Numerous pores remain in the same location along the ridges, but some of their shapes vary. The oval area is further complicated by additional vertical creases in the upper right of Fig. 10a. The noted differences could not be reconciled due to lack of additional images of sufficient clarity and/or area; the data were insufficient to confirm or refute the possibility of transient but reversible changes in pores suggested by the physiological studies described in the Background section.

Incipient ridges were neither permanent nor persistent. In 148 fingers examined, approximately 40% had incipient ridges. No incipient ridges appeared or disappeared, but 82% differed in appearance over time. Although some incipient ridges in Fig. 10c and d were present in both images, there were substantial differences in incipient ridges within the oval area.

Neither H1 nor H2 was fully supported in observations of level 3 details in the skin over a short term.

4.3.6. Level 3 variations over a long term (plus incipient ridges)

Edge shape was neither permanent nor persistent in the long-term. In Fig. 11b, the ridges have become rough and scaly after the passage of eight years, especially noticeable below the core. In addition, edge shape changes were observed in Fig. 11c, d and e. Ridge width was neither permanent nor persistent

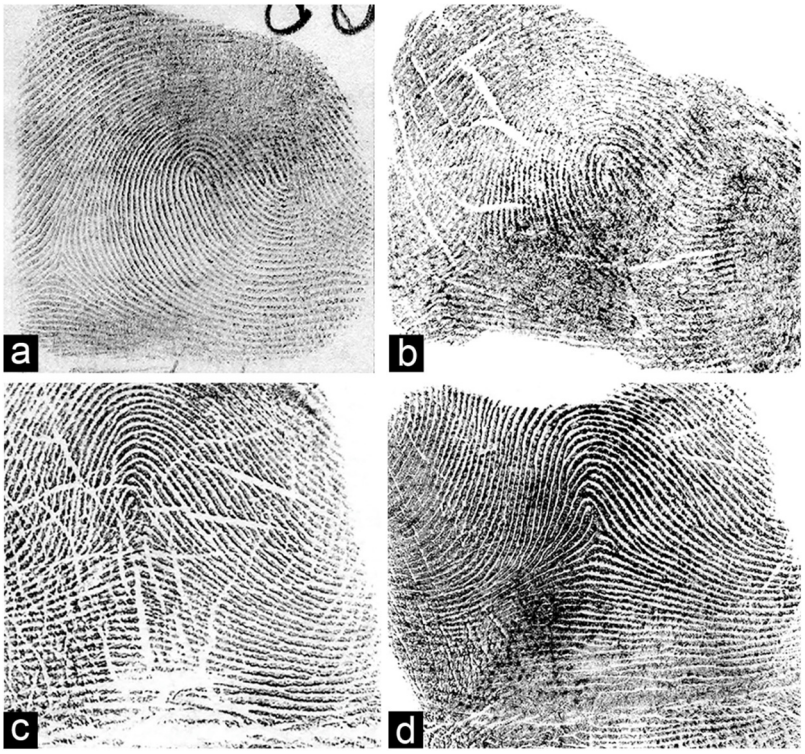


Fig. 14. Persistence of level 1 detail of ink rolled impressions over a long term: (a,b) mates at ages 19 and 62; (c,d) mates at ages 23 and 50. White lines appeared (a,b) and disappeared (c,d) at the same rate, approximately 44%. Due to the areas recorded, ridge flow could be misinterpreted (a,b).

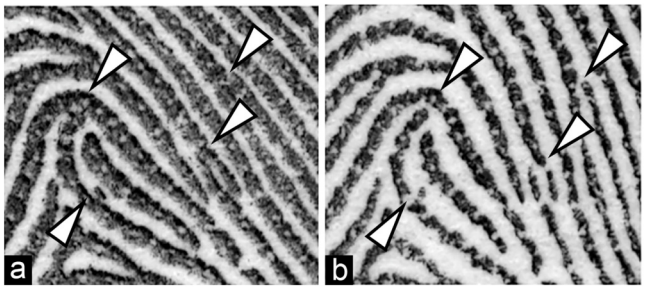


Fig. 15. Persistence of level 2 detail of ink rolled impressions over a short term: (a,b) mates separated by a 1-month interval at age 31. A dividing ridge in (a) appears as an

(Fig. 11c, d and e). Due to high variability observed in the short term, ridge spacing in the long term defaults to being persistent but not permanent.

Pores were neither permanent nor persistent. Although some pores appeared similar over the long term, there were variations in their appearance and attributes associated with them, e.g., size and shape (Fig. 7). Ridge breaks change the overall appearance and visibility of the pores.

Incipient ridges were neither permanent nor persistent. In 32 finger pairs examined over a long term, seven had incipient ridges, one of which differed in appearance. In Fig. 11d and e (contemporaneous), the marked incipient ridge appears distinctly separated and might even be considered a short, thin ridge. Ten years prior, that same ridge looks much thinner and longer, almost fused with the main ridge (Fig. 11c).

Neither H1 nor H2 was fully supported in observations of level 3 details in the skin over a long term.

4.4. Appearance of impressions over time

A friction ridge impression is produced when the three dimensional palmar surface of a finger is pressed or rolled against a surface to create a two dimensional image. Distortions in impressions arise due to differences in pressure, movement, force, and matrix. Pressure can affect all levels of detail. Resulting ridge width and shape, contributing to the overall friction ridge arrangement, may vary. With increasing pressure there are numerous effects on impressions, including: widening of ridges and smoothing of ridge edges, narrowing of furrows, variation relative to pattern type (loop or whorl), increased visibility of dots and incipient ridges, a ridge ending touching an adjacent ridge, and more pronounced effects in the finger pad area than at the distal end [8,79,151,154]. In recognition of these effects of pressure, which may be compounded by moisture and over- or under inking, Interpol recommends that differences between ending and dividing ridges in impressions "are within normal tolerances and have no fundamental relevance" [133, p. 15]. Torque, and the direction it is applied, may cause other effects [79,151,155]. Live scan impressions were not analyzed for this phase of the study because there was only a single capture and live scans often failed to show the degree of detail, particularly level 3, revealed by the other methods.

4.4.1. Level 1 variations over a short term (plus cuts, scars, and white lines)

Persistence and transfer fidelity of level 1 detail were observed over a short-term period. Ridge flow is affected by the pressure and/or movement applied to a finger while it is being recorded. For example, the core appears more elongated in a May ink rolled image (Fig. 12b) than in the April one (Fig. 12a). The effect is even more pronounced for the June (Fig. 12d) and April (Fig. 12c)

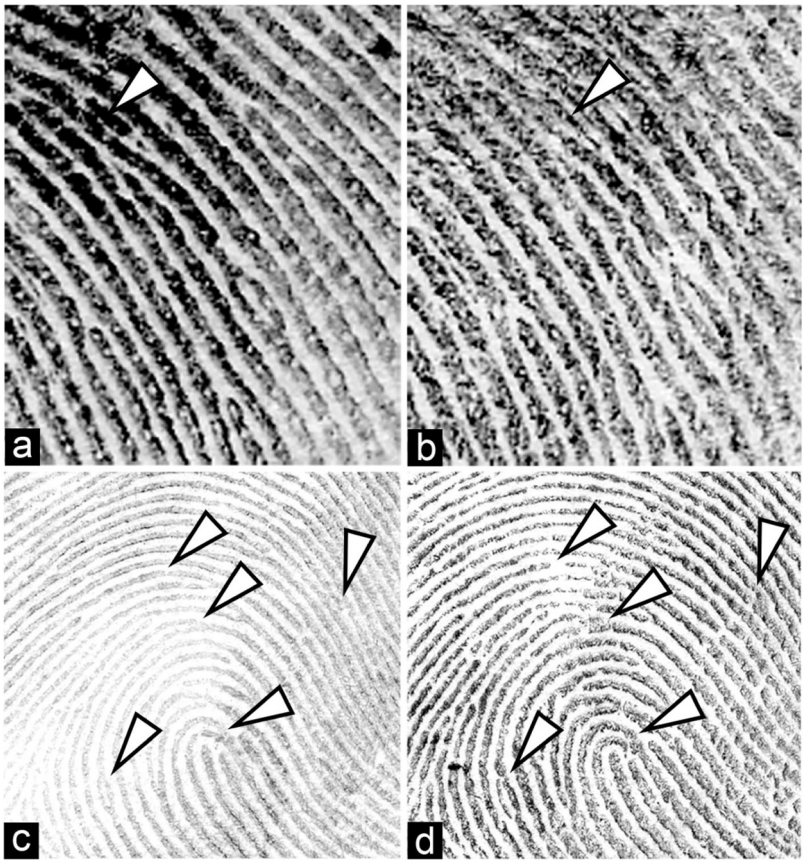


Fig. 16. Persistence of level 2 detail of ink rolled impressions over a long term: (a,b) mates at ages 21 and 58; (c,d) mates at ages 23 and 56. A short ridge in (a) more resembles an incipient ridge in (b). Compared to (c), (d) has more ridge breaks; an enclosure was disrupted (bottom arrow); bifurcations were broken (arrows to the right of and near the core); a large ridge gap (topmost arrow); and an ending to dividing ridge (rightmost arrow).

holographic images of the same finger. Similar distortions, especially in the core area, were readily attained in supplementary experiments (data not shown). Note that core elongation was also observed in the skin, so therefore may be partially independent of distortion (Section 4.3.2). Variations in appearance of level 1 may arise due to the physiological occurrences mentioned above, the pliability of the skin, the variety of pressure and movement involved in some recordings, the presence of cuts, creases, or white lines, and intermittent artifacts attributable to capture method.

Fig. 13 is representative of a crease/white line varying over a short time. Fewer creases appear in April than in June. The April photo (not shown), ink rolled, and holographic image all show a horizontal crease/white line across the top of the core. The June photo, ink rolled, and holographic image show that crease/white line as well as a second crease/white line below it.

Both H1 and H2 were supported in observations of level 1 details in impressions over a short term.

4.4.2. Level 1 variations over a long term (plus cuts, scars, and white lines)

Level 1 detail was persistent over a long-term period. White lines can be useful for orientation and searching, but due to observed variability, are not recommended for comparisons. In a sample of 158 paired impressions, 89% had white lines. The rate at which white lines appeared or disappeared over time was approximately the same (not significantly different, $p > 0.05$). In 42% of the cases, white lines appeared in the recent print, sometimes to the detriment of quality (Fig. 14b). Over a long term, white lines disappeared 46% of the time (Fig. 14d). In 13% of the cases, white lines appeared and disappeared between the paired prints. The upper pair of images is noteworthy because the overall ridge flow may appear different—from a clear double loop whorl (Fig. 14a) to what might at first glance look to be a simple loop in the impression taken 43 years later (Fig. 14b), due to the fact that the later impression was not fully recorded. Scars, sometimes surrounded by pinched skin, were consistently recorded.

Both H1 and H2 were supported in observations of level 1 details in impressions over a long term.

4.4.3. Level 2 variations over a short term

Level 2 detail was persistent over a short-term period. Pressure affects the appearance of level 2 detail. Occasionally, dividing ridges appeared as ending ridges and ending ridges appeared as dividing ridges. A spur in the April ink rolled impression (Fig. 15a) appears as separate ridges in the fingerprint acquired one month later (Fig. 15b). A dividing ridge in the core area of Fig. 15a appears as an ending ridge in Fig. 15b. The ending ridge six ridges to the right of the core in Fig. 15b could be interpreted as a dividing ridge in Fig. 15a. All three areas in the April ink rolled are consistent with the April and June photographs (not shown).

Both H1 and H2 were supported in observations of level 2 details in impressions over a short term.

4.4.4. Level 2 variations over a long term

Level 2 detail was persistent over a long-term period. Variations observed were attributable to pressure, inking, movement, and

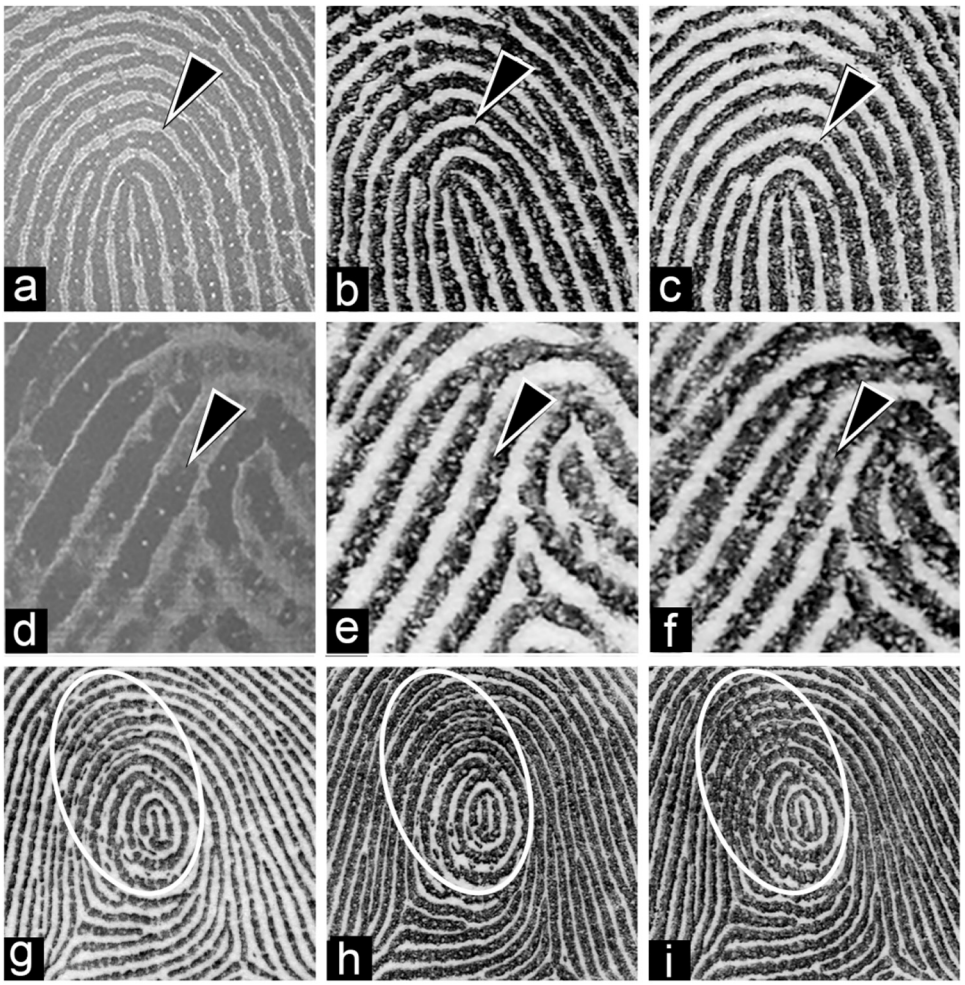


Fig. 17. Lack of persistence of level 3 impressions over a short term: (a,b,c) mated ink rolled impressions collected in May, June, and August at age 26 show persistence of edge shape and ridge width (arrows); (d,e,f) mated ink rolled impressions collected in April, June, and August at age 31 exhibit variations in pore appearance (arrows); (g,h,i) mated impressions at age 27, where circled areas show variation in incipient ridges (April rolled ink, June inked flat, August inked flat, respectively).

changes in pliability of the skin. In Fig. 16a, a short ridge more closely resembles an incipient ridge in the impression collected 37 years later (Fig. 16b). Multiple ridge breaks distinguish the recent impression (Fig. 16d) from that taken 33 years prior (Fig. 16c). Moving from the bottom of the image pair, an enclosure in Fig. 16c has become disrupted by two ridge breaks in Fig. 16d. A bifurcation at the core is disrupted by one ridge break in Fig. 16d. Another bifurcation above the core is disrupted by two ridge breaks in Fig. 16d. The topmost pointer shows a large ridge gap in Fig. 16d. The pointer at the extreme right highlights an ending ridge in Fig. 16c that appears as a dividing ridge in Fig. 16d.

Both H1 and H2 were supported in observations of level 2 details in impressions over a long term.

4.4.5. Level 3 variations over a short term (plus incipient ridges)

Ridge width and edge shape were persistent in impressions over a short-term period, despite some variations. For example, ridge and/or furrow width may vary from one impression to another, either within specific areas or globally. Having observed ridge spacing variations in skin over a period of minutes, no conclusions were drawn for impressions. Ridge edges are not always smooth but have protuberances that make them uneven. Figs. 17a–c highlight a ridge protuberance that is persistent from May to August. Depending on whether or not those protuberances come in contact with the surface, or if ink is not applied evenly, as well as the degree of pressure applied, they may or may not appear from one impression to the next. By analogy, if one considers making a two-dimensional impression of a rubber ball, different degrees of pressure could create impressions ranging from a mere point to the diameter of the ball. The finger pad is like an elastic sphere, with increased pressure producing a greater surface area of contact until a limit is reached [151,156].

Pore presence was not persistent. Pore location, shape, and size vary depending on the capture technique and pressure applied. Since variations in pores were observed in photographs of the skin, and since other investigators have reported that pore appearance can change over time, the inconsistencies observed in impressions showing pores is not surprising. Four equally spaced pores, located two ridges left of the delta, are clearly present in the April holographic image (Fig. 17d), and June ink rolled (Fig. 17e). However, in the August ink rolled image (Fig. 17f) the four pores are obscured, due either to excess pressure or to physiologic variation. Referring to Fig. 15, most of the pores are present in both impressions but they vary in shape and size. However, the pore to the left of the dot in the core area (leftmost arrow) is larger in the May ink rolled (Fig. 15b). It is unknown whether all these variations in impressions are more likely due to excess pressure and ink than to a change over time.

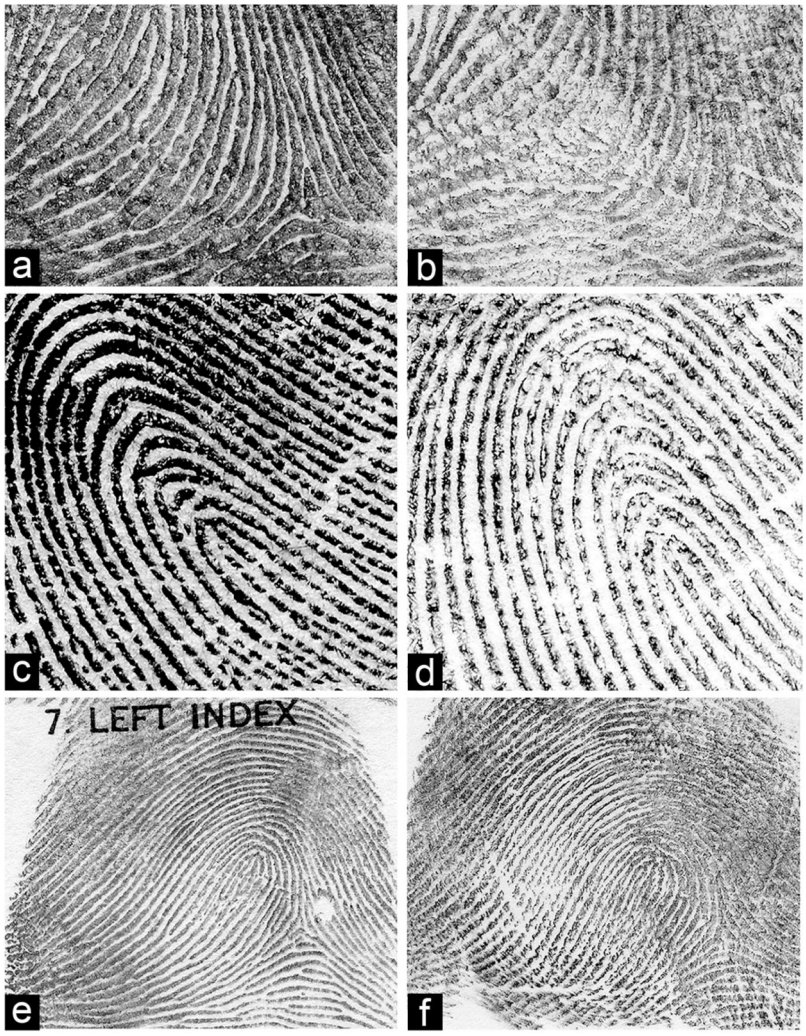


Fig. 18. Lack of persistence of level 3 impressions over a long term: (a,b) variations in edge shape and pore appearance between ages 45 and 59; (c,d) variation in pore size between ages 19 and 48; (e,f) variation in incipient ridges between ages 18 and 71.

Incipient or immature ridges were not persistent. Their overall absence or presence in different recorded impressions of the same finger tends to be similar, but there were attributes about them that change: number, size, length, width, etc. Incipient ridges are thinner, shallower, and often shorter than fully developed ridges, consequently they tend to be even more susceptible to variation in impressions, as illustrated in Fig. 17g–i. For this reason, comparison counts of incipient ridges were not attempted for impressions.

Neither H1 nor H2 was fully supported in observations of level 3 details in impressions over a short term.

4.4.6. Level 3 variations over a long term (plus incipient ridges)

With the exception of ridge width, level 3 details were not persistent over a long-term period in impressions (as well as being inextricably affected by pressure, inking, movement, capture method, etc.). Edge shape was not persistent, nor were pores visible in an impression taken 14 years after the first (Fig. 18b). In Fig. 18c and d, while many pores were persistent in location after an elapsed time of 29 years, they are not necessarily persistent in size. Fig. 18f illustrates incipient ridges that appeared after an interval of 53 years.

Neither H1 nor H2 was fully supported in observations of level 3 details in impressions over a long term.

5. Conclusions

The presumption of persistence, which underlies the practice of comparison and identification of fingerprints, was upheld. All photographs and all impressions of the same finger, irrespective of the length of time separating their capture, would be identifiable as originating from the same source.

For all recording techniques, image quality was a key factor in the ability to assess permanence and persistence. The extent to which features are faithfully transferred is directly linked to the technique used to capture the image, especially for level 3 detail. Each technique used to record the fingerprint images has inherent limitations in quality relating to the technique itself as well as its application. Due to their frequently marginal quality, many of the available photos did not adequately represent ground truth.

The participating fingerprint examiners conducted every comparison with the deliberate mindset of attempting to detect dissimilarities or discrepancies [14], however slight. The variations noted were rarities within an overwhelming similarity of features.

As defined within this study, a single variation makes any attribute not permanent. The operational manifestation of permanence for fingerprint comparison is the concept of persistence. Thus, the threshold for persistence is relaxed to accommodate physiological and image capture factors.

Neither hypothesis was fully supported for all levels of detail (i.e., remaining permanent on friction ridge skin and persistent on skin and in impressions).

Level 1 detail was permanent in the skin and persistent in impressions.

Level 2 detail in the skin and in impressions was persistent. However, though permanent in the short term, level 2 detail was not permanent in the long term. Notably, the small changes observed were only in appearance; there were no changes in the presence of new, or absence of existing, minutiae. The differences we saw in skin were so small, rare, and insignificant that they would be unconditionally considered persistent in an operational exam.

Level 3 detail, with few exceptions, was neither permanent nor persistent. Ridge width was both permanent and persistent. Ridge edge shape, though permanent and persistent in the short term, was neither permanent nor persistent in the long term. Since ridge spacing was persistent but not permanent due to observed variability between photographs taken within minutes of one another. Pore presence and incipient ridges were neither permanent nor persistent.

This study reinforces the prevailing use of the term persistence, rather than permanence, to describe the sustained existence of friction ridge detail that enables a trained examiner to identify two impressions or photographs as being made by the same finger, even if the two images were captured over varying lengths of time. The nature of the dissimilarities noted in this study have been recognized from the first use of fingerprints as a means of identification. The apparent differences observed are of the same level of operational significance as those noted by Galton, for which "an exact correspondence between the details of two minutiae is of secondary importance" and against which he offered "caution against attaching undue importance to disaccord in the details of the minutiae that are found in the same place in different prints" [41, pp. 91–92].

CRediT authorship contribution statement

Keith L. Monson: Data curation, Formal analysis, Investigation, Methodology, Writing - original draft, Writing - review & editing. **Maria Antonia Roberts:** Data curation, Formal analysis, Investigation, Methodology, Writing - original draft, Writing - review & editing. **Kathryn B. Knorr:** Formal analysis, Validation, Writing - review & editing. **Sherine Ali:** Formal analysis, Validation, Writing - review & editing. **Stephen B. Meagher:** Investigation. **Kevin Biggs:** Investigation. **Patti Blume:** Investigation. **Donna Brandelli:** Investigation. **Albert Marzioli:** Investigation. **Robert Reneau:** Investigation. **Frank Tarasi:** Investigation.

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Keith L. Monson: Data curation, Formal analysis, Investigation, Methodology, Writing - original draft, Writing - review & editing. **Maria Antonia Roberts**: Data curation, Formal analysis, Investigation, Methodology, Writing - original draft, Writing - review & editing. **Kathryn B. Knorr**: Formal analysis, Validation, Writing - review & editing. **Sherine Ali**: Formal analysis, Validation, Writing - review & editing. **Stephen B. Meagher**: Investigation. **Kevin Biggs**: Investigation. **Patti Blume**: Investigation. **Donna Brandelli**: Investigation. **Albert Marzioli**: Investigation. **Robert Reneau**: Investigation. **Frank Tarasi**: Investigation.

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Appendix A. Supplementary data

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